

Smart Reward Point Management and Redemption System

For Student Performance Evaluation

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The Smart Reward Point Management and Redemption System is a web-based platform developed to automate and manage reward points earned by students through academic and extracurricular activities. The system ensures transparency, efficiency, and motivation among students by tracking their performance and rewarding their participation across multiple categories of engagement.

With this revision, the system is expanded to include a dedicated Teacher role, positioned between the Admin and Student roles, so that domain-specific activities — hackathons, problem-solving practice, cultural events, internships, and certifications — can be managed by faculty members responsible for those categories, while the Admin retains full oversight of the entire platform.

1.2 Intended Audience

- Project guides, evaluators, and academic reviewers assessing the system design.
- Development team members implementing the frontend, backend, and database layers.
- Institutional stakeholders (administration and faculty) who will operate the system.

2. Objectives

- Automate reward point allocation, deduction, and tracking across all student activities.
- Introduce a Teacher role to decentralize activity management by domain/category, reducing Admin workload.
- Improve transparency in student performance tracking through a unified dashboard and ranking system.
- Encourage participation through gamified rewards, leaderboards, and badges.
- Provide a centralized system for academic, co-curricular, and extra-curricular activity data.
- Enable effective, role-appropriate communication between students, teachers, and administrators.
- Provide real-time notifications and actionable performance insights.
- Enforce role-based access control so that each user interacts only with data relevant to their role.

3. Scope

This system is designed for use within a college environment and supports three categories of users: Admin, Teacher, and Student. It includes:

- Reward Point (RP) allocation, deduction, and history tracking.
- Category-based activity management (hackathons, coding contests, competitions, PS activities, cultural events, internships, and NPTEL certifications).
- Role-based access control distinguishing Admin, Teacher, and Student permissions.
- Student performance monitoring combining RP and internal marks.
- Query interaction system between students and the teacher responsible for their category.
- Ranking, leaderboard, and analytics features.
- Automated, role-targeted notifications.

The system replaces manual, spreadsheet-based tracking with a digital, scalable, and role-segmented solution suitable for institutional deployment.

Out of scope: payroll or financial disbursement of rewards, integration with external national certification portals (e.g., NPTEL's own verification API), and mobile native applications in the initial release.

4. System Overview

The system is composed of three cooperating modules aligned to the three user roles:

- **Admin Module** — full-system oversight: manages Teacher and Student accounts, monitors all activities and RP transactions, and reviews reports and analytics.
- **Teacher Module**— category-scoped management: Faculty 1 (Hackathons, Coding Contests, Competitions), Faculty 2 (PS Activities), and Faculty 3 (Cultural Events, Internships, NPTEL Certifications) each post activities, assign RP, and respond to queries strictly within their assigned category.
- **Student Module** — self-service access: dashboard, ranking, RP history, internal marks, and the ability to raise queries.

The diagram below illustrates the original system flow between Admin and Student roles through login, RP allocation, notifications, and the central database. This flow now extends through the Teacher role described in the following sections.

5. User Roles and Access Control

The system enforces three distinct roles, each with a defined access boundary. Role-based access control (RBAC) is implemented using JWT-based authentication, so that every API request is validated against the caller's role and, for teachers, their assigned category.

5.1 Role Summary

Role	Access Level	Key Permissions
Admin	Full system access	Manage Teacher and Student accounts; monitor all activities and RP; view system-wide reports and analytics; respond to escalated queries.
Teacher	Limited, category-scoped access	Manage activities only within assigned category; post activities; assign RP; view limited (category-relevant) student data; respond to related student queries.
Student	Self-service access	Login with email; view dashboard and ranking; participate in activities; view RP history and internal marks; raise queries.

5.2 Teacher Categories

The Teacher role is subdivided into three category types, each scoped to a distinct set of activities. A teacher account can only create activities, assign RP, and view queries that fall within its assigned category.

Teacher Category	Assigned Activity Domains
Faculty 1	Hackathon, Coding Contests, Competitions
Faculty 2	Problem-Solving (PS) Activities
Faculty 3	Cultural Events, Internships, Certifications (NPTEL)

6. Functional Requirements

6.1 Admin Module

Admin Login

- The system shall provide secure login with JWT-based authentication.

Teacher and Student Management

- The Admin shall be able to create, update, deactivate, and remove Teacher and Student accounts.
- The Admin shall assign each Teacher account to one of the three faculty categories.

Query Oversight

- The Admin shall be able to view all student queries and escalate unresolved ones.
- The Admin shall track query status (resolved / pending).

Activity and RP Monitoring

- The Admin shall view all activities and RP transactions posted by any teacher.
- The Admin shall be able to override or correct RP entries when required.

Student Reports

- The Admin shall view a list of students and, per student, their total RP, RP history, activities participated in, internal marks, and a performance summary.

6.2 Teacher Module

Teacher Login

- The system shall authenticate the teacher and load only their assigned category context.

Category-Based Activity Posting

- The teacher shall create and post activities restricted to their assigned category (Faculty 1, 2, or 3).
- The teacher shall define the RP value associated with each activity.

RP Assignment

- The teacher shall assign RP to students who complete or participate in an activity within their category.
- The teacher shall be able to deduct RP with a mandatory reason where policy requires.

Student Data Visibility

- The teacher shall view participation and RP data only for students engaged with their category's activities.

Query Handling

- The teacher shall view and respond to queries raised by students that relate to their category.

6.3 Student Module**Student Login**

- The student shall log in using their institutional email credentials.

Dashboard with Ranking

- The dashboard shall display total RP, average RP, RP status (Low / Good), overall ranking, and personalized recommendations for earning additional RP.

Activities Section

- The student shall view activities posted by any teacher, filterable by category.
- The student shall view the RP value associated with each activity before participating.

Rewards History

- The student shall view a complete RP history, distinguishing RP earned from RP used or deducted.

Internal Marks Module

- The system shall display internal marks and show how they are derived from accumulated RP.
- The system shall indicate the additional RP required to reach the next internal-marks tier.

Student Queries

- The student shall submit queries, which are routed to the teacher responsible for the relevant category (or to Admin if uncategorized).
- The student shall view responses to previously raised queries.

7. Non-Functional Requirements**7.1 Security**

- Role-based authentication and authorization using JWT.
- Passwords stored using a strong one-way hashing algorithm (e.g., bcrypt).
- Category-level authorization checks enforced on every teacher-scoped API endpoint.

7.2 Performance

- Dashboard and ranking queries shall return within 2 seconds under normal load.
- Efficient, indexed database queries for RP aggregation and ranking computation.

7.3 Usability

- Simple, intuitive UI tailored to each role's permitted actions.
- Responsive, mobile-friendly design.

7.4 Scalability

- The system shall support a growing number of students, teachers, and activities without redesign.

7.5 Reliability

- Accurate, auditable RP tracking with a complete transaction log.
- High system availability, with graceful handling of notification-service downtime.

7.6 Maintainability

- Modular architecture separating Authentication, Role Modules, Activity, RP, Notification, and Reporting concerns, so that new teacher categories or activity types can be added with minimal change.

8. System Architecture

The system follows a layered, three-tier architecture:

- Presentation Layer — React.js single-page application, rendering role-specific views (Admin, Teacher, Student) after JWT-based role resolution.
- Application / API Layer — Node.js with Express.js, exposing REST endpoints grouped by module (Auth, Admin, Teacher, Student, Activity, RP, Query, Notification, Reports). Middleware validates JWT claims for role and, where applicable, teacher category, before any request reaches business logic.
- Data Layer — MySQL relational database storing users, activities, RP transactions, queries, and notifications, with foreign-key constraints enforcing referential integrity between entities.

Cross-cutting services include a Notification Service (Firebase Cloud Messaging / email) triggered by RP changes and query responses, and a Reports & Analytics component that aggregates RP and activity data for dashboards and rankings.

8.1 Technology Stack

Layer	Technology
Frontend	React.js
Backend	Node.js with Express.js
Database	MySQL
Authentication	JWT
Notifications	Firebase Cloud Messaging / Email
Deployment	Frontend: Vercel; Backend: Render; Database: MySQL
DevOps Tools	Git & GitHub

9. Module Descriptions

9.1 Authentication Module

Handles login for all three roles, issues JWTs encoding role and (for teachers) category, and enforces session validity.

9.2 Admin Module

Provides full administrative control: account management, system-wide monitoring, and access to all reports.

9.3 Teacher Module (New)

Provides category-scoped tools for posting activities, assigning RP, viewing relevant student data, and responding to queries.

9.4 Student Module

Provides the student-facing dashboard, activity browsing, RP history, internal marks view, and query submission.

9.5 Activity Management Module

Manages creation, categorization, and lifecycle of activities posted by teachers, with category tagging (Faculty 1 / 2 / 3).

9.6 Reward Points (RP) Management Module

Handles RP allocation, deduction, and transaction history; maintains an auditable log of every RP change and the teacher who authorized it.

9.7 Dashboard & Ranking Module

Computes and displays total/average RP, RP status, and student rankings, updated whenever RP changes.

9.8 Internal Marks Calculation Module

Converts accumulated RP into internal marks using a defined conversion policy, and shows students the RP needed for improvement.

9.9 Query/Communication Module

Routes student queries to the appropriate teacher category (or Admin) and tracks resolution status.

9.10 Notification Module

Delivers real-time alerts for RP additions/deductions, low-RP status, new activities, and query responses.

9.11 Reports & Analytics Module

Aggregates activity, RP, and performance data into reports for Admin review and institutional analytics.

9.12 Additional Platform Features

- Leaderboard system displaying top-performing students institution-wide and per category.
- Badge system (Gold / Silver / Bronze) awarded based on RP milestones.
- Smart recommendations suggesting activities likely to improve a student's RP status.
- Category-based activity filtering across the Teacher and Student modules.

10. Workflow Diagrams

The following diagrams illustrate the step-by-step system flow for each role, including how activity participation flows into RP allocation and downstream notifications.

10.1 Admin Workflow

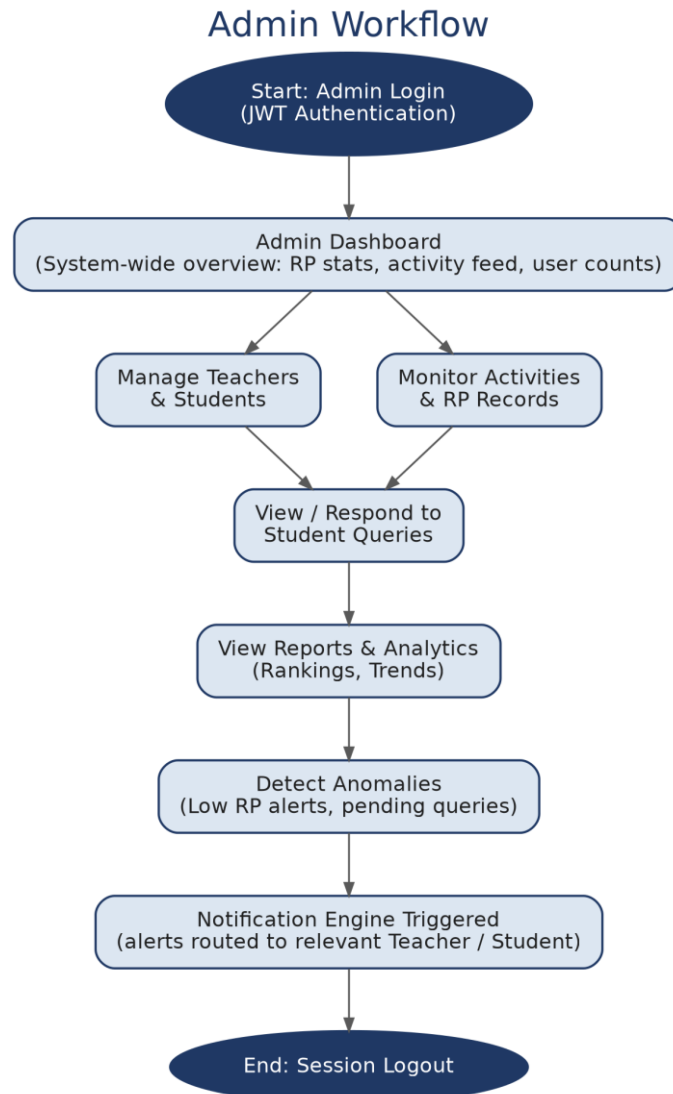


Figure 10.1 — Admin workflow: from login through monitoring, reporting, and notification triggering.

10.2 Teacher Workflow

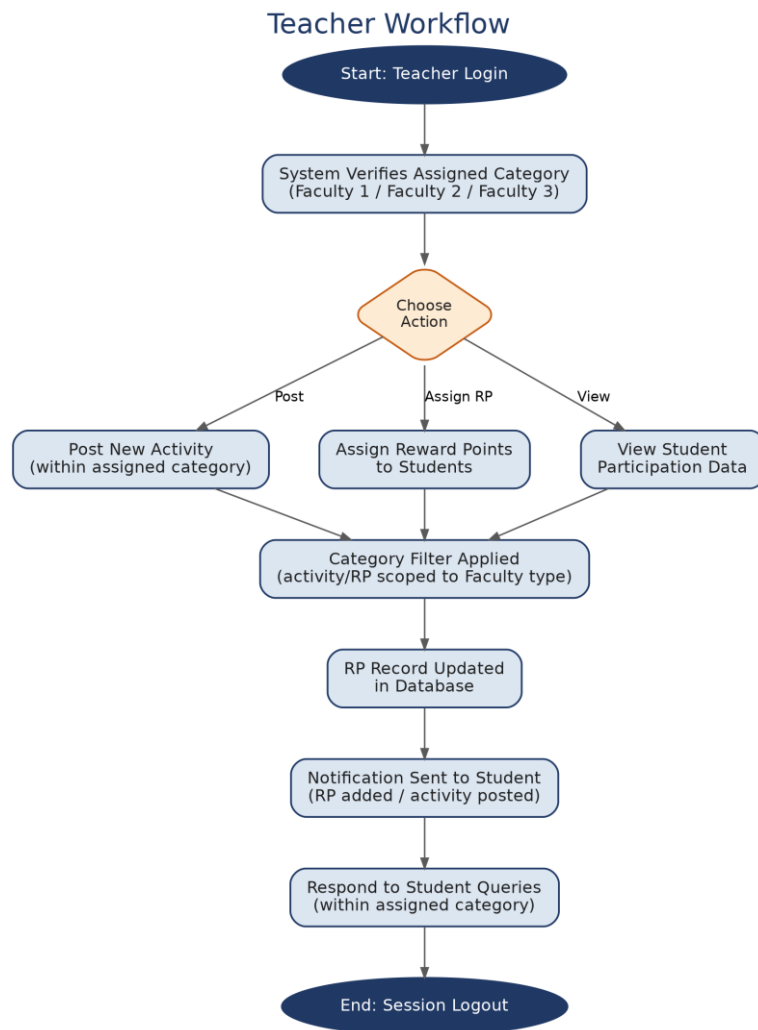


Figure 10.2 — Teacher workflow: category verification, activity posting, RP assignment, and query handling.

10.3 Student Workflow

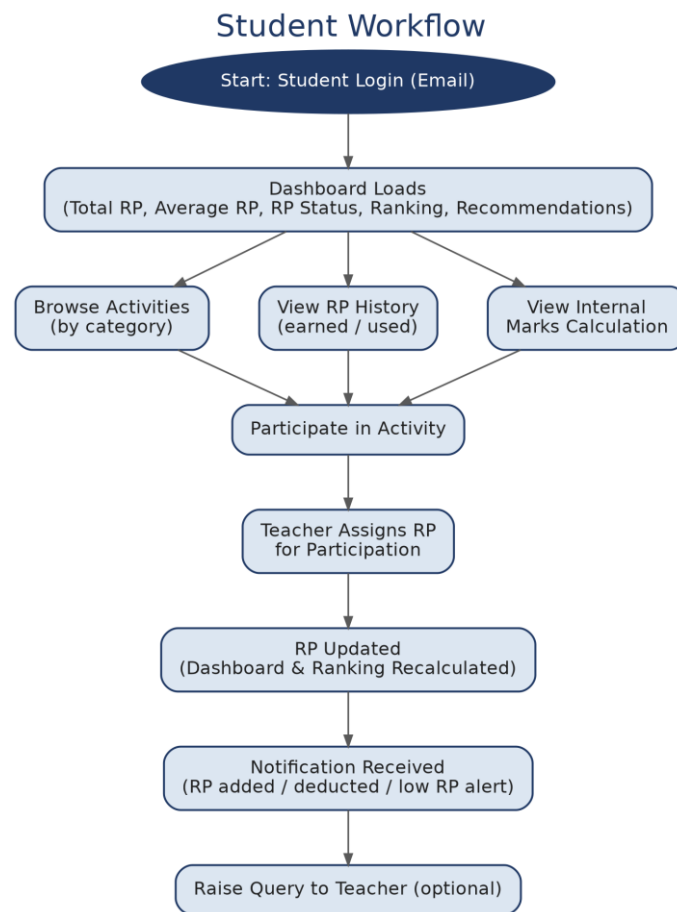


Figure 10.3 — Student workflow: dashboard access, activity participation, RP update, and notification receipt.

11. ER Diagram

The entity-relationship model below defines seven core entities — Admin, Teacher, Student, Activity, RewardPoints, Query, and Notification — along with their primary keys, foreign keys, and relationships.

Entity-Relationship (ER) Diagram

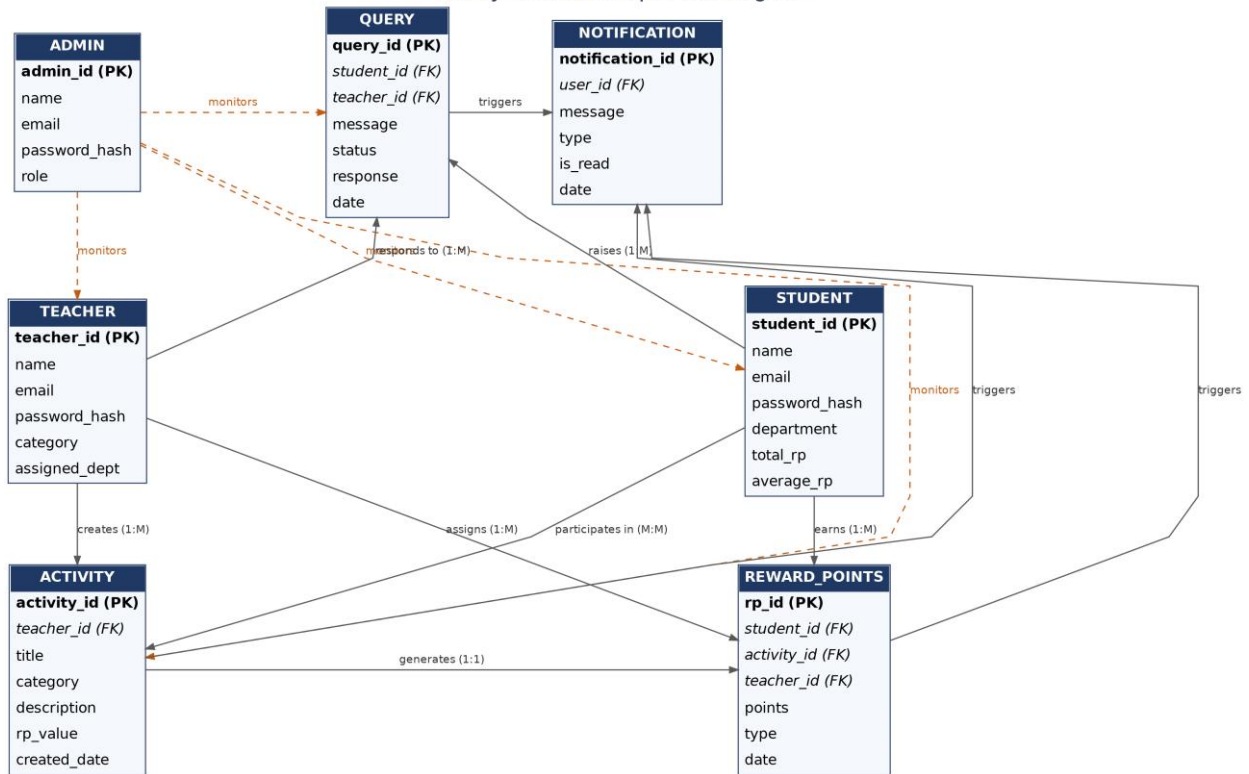


Figure 11.1 — Entity-Relationship diagram for the Smart Reward Point System.

11.1 Entity Reference

Entity	Primary Key	Foreign Key(s)	Relationship Notes
Admin	admin_id (PK)	—	Monitors all entities system-wide.
Teacher	teacher_id (PK)	—	Creates Activity; assigns RewardPoints; responds to Query.
Student	student_id (PK)	—	Participates in Activity; earns RewardPoints; raises Query.
Activity	activity_id (PK)	teacher_id (FK)	Created by Teacher; generates RewardPoints on completion.
RewardPoints	rp_id (PK)	student_id, activity_id, teacher_id (FK)	Links Student, Activity, and Teacher; triggers Notification.
Query	query_id (PK)	student_id, teacher_id (FK)	Raised by Student; answered by Teacher; triggers Notification.
Notification	notification_id (PK)	user_id (FK)	Generated by Activity, RewardPoints, or Query events.

12. UI Wireframes and Screen Mockups

This section presents low-fidelity wireframes for the primary screens of the Admin, Teacher, and Student modules. Each wireframe is a structural layout reference using placeholder blocks in place of final text, imagery, and iconography; it defines page regions, navigation, and interactive controls rather than final visual styling. The wireframes correspond directly to the functional requirements defined in Section 6.

12.1 Admin Module Screens

Student Roster and Reports Screen

This screen supports the Admin's Student Reports function (Section 6.1). The left navigation panel provides access to the Admin's management areas, with the current section highlighted. The top utility bar offers a keyword search, quick-filter access, notification and settings icons, and the logged-in Admin's profile. A summary notice panel in the upper-right calls out items requiring attention. Below it, a row of four overview cards presents institution-wide metrics such as total students, active activities, pending queries, and RP issued. The main panel is split into two areas: a student roster table on the left, listing each student with selectable rows for drilling into individual RP history, activities, and internal marks; and a reports panel on the right, listing generated report entries with a status indicator and an action button for each. The layout allows the Admin to move from a system-wide overview to individual student detail without leaving the page.

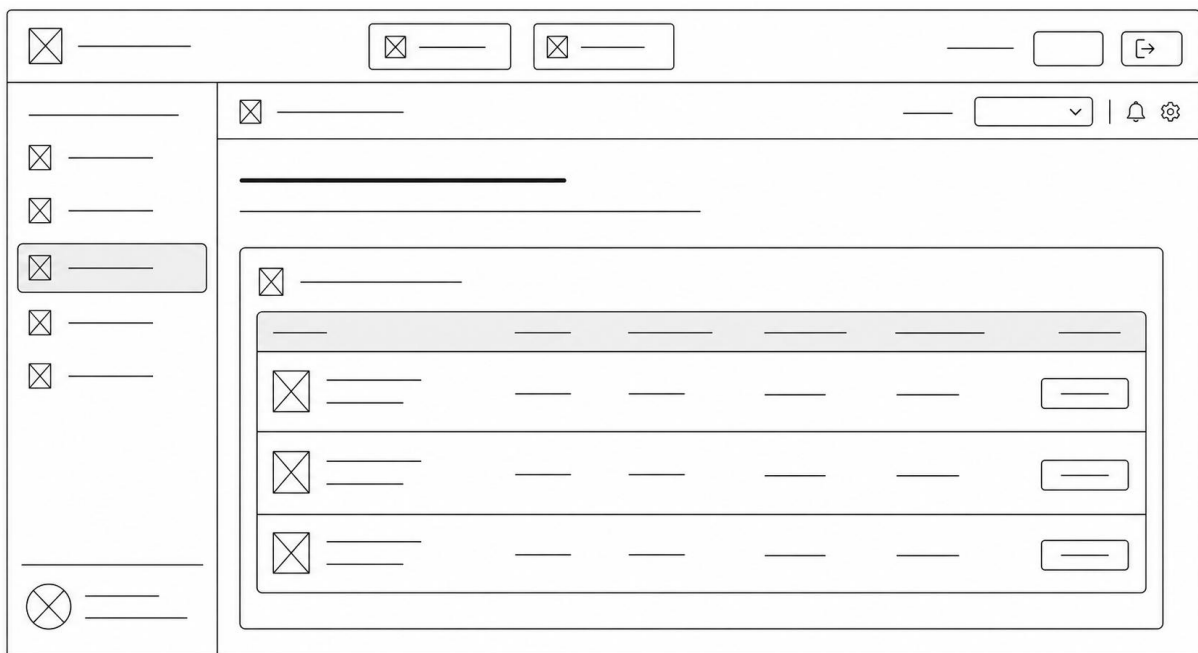


Figure 12.1 — Student Roster and Reports Screen (Admin).

12.2 Teacher Module Screens

Activity Posting and RP Allocation Screen

This screen supports the Teacher's Category-Based Activity Posting and RP Assignment functions (Section 6.2). The top bar retains global navigation and category context, while the left sidebar lists the teacher's available actions within their assigned category. The main panel is split into two forms presented side by side. The left form is used to create a new activity: it captures the activity title, a description in a multi-line

text area, and an RP value, with a primary action button to post the activity. The right form is used to assign RP to a student for an existing activity: a segmented control switches between assignment modes, two dropdown selectors identify the target student and activity, and two action buttons confirm or cancel the assignment. Presenting both forms together lets a teacher post a new activity and immediately allocate RP for it within a single screen.

The wireframe illustrates a web interface for teachers. It features a top navigation bar with a close button (X), two checkboxes, and a search bar. The main content area is divided into three sections. On the left is a sidebar with a list of items, each preceded by a checkbox, and a close button at the bottom. The central area contains a form for posting a new activity, including a title, a description text area, and a response area. The right section contains a form for allocating RP, with a segmented control for assignment modes, two dropdown menus for student and activity selection, and two action buttons.

Figure 12.2 — Activity Posting and RP Allocation Screen (Teacher).

Query Management Screen

This screen supports the Teacher's Query Handling function (Section 6.2) and feeds into the Admin's Query Oversight function (Section 6.1) for escalated items. The left sidebar lists the teacher's category-scoped sections, with the queries section highlighted. The main panel lists incoming student queries as stacked cards, each showing the query subject line, submitting student, and category tag, together with a response action button. Expanding a card reveals the associated response area, shown as a shaded text block, where the teacher can view or enter a reply. This layout allows a teacher to review and respond to every query relevant to their category from a single scrollable list.



Figure 12.3 — Query Management Screen (Teacher).

12.3 Student Module Screens

Student Dashboard Screen

This screen supports the Student's Dashboard with Ranking function (Section 6.3). The top bar provides quick links to the Dashboard and Leaderboard, alongside a search field, notifications, and settings. The left sidebar lists the student's available sections, with the dashboard entry highlighted. The main panel opens with a page title and summary line, followed by two side-by-side statistic cards presenting the student's total RP and average RP. Below the cards, a highlighted status panel displays the student's current RP status and a progress indicator toward the next milestone. A further card beneath presents personalized recommendations for earning additional RP. The layout is intended to give the student an immediate, at-a-glance view of their standing on login.

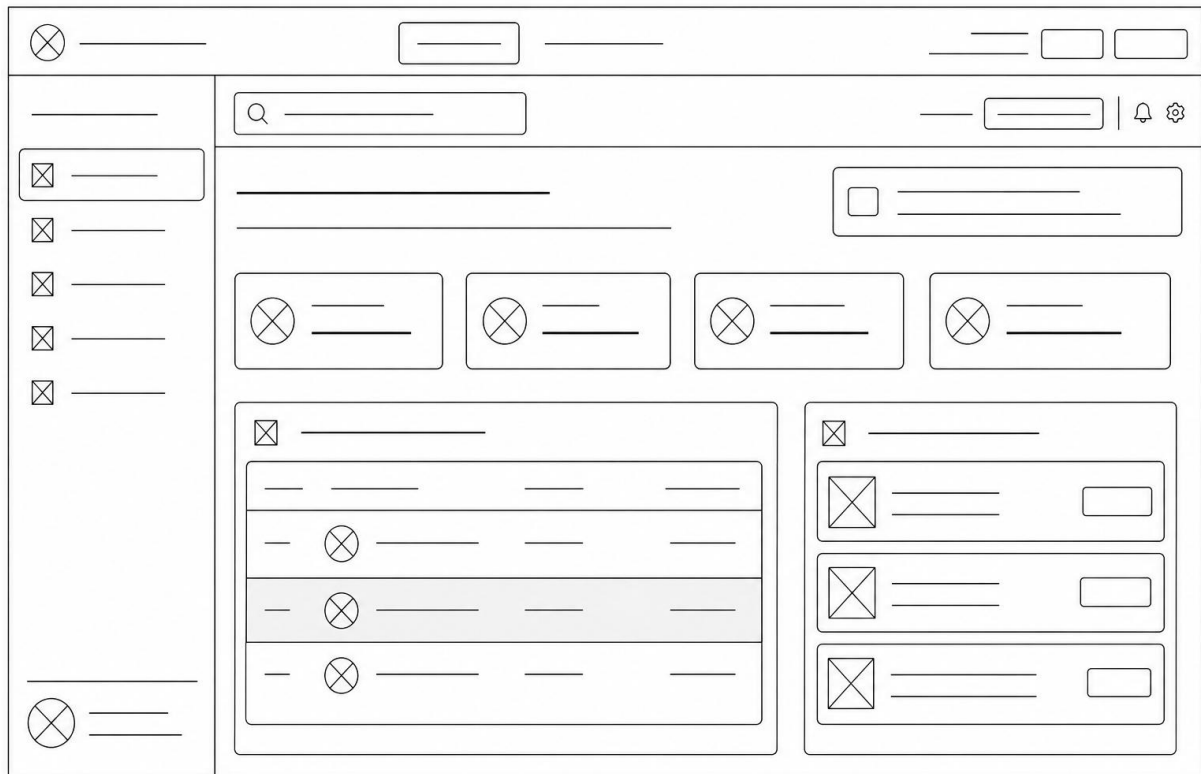


Figure 12.4 — Student Dashboard Screen.

Available Activities Screen

This screen supports the Student's Activities Section function (Section 6.3). A search field at the top of the main panel allows the student to locate specific activities, while a status filter in the upper-right narrows the list by participation state. Activities are presented as a row of cards, each displaying the activity title, its category, a short description, and a call-to-action button for viewing details or participating. The left sidebar retains navigation to the student's other sections, with the activities entry highlighted. This card-based layout allows the student to browse and filter activities posted by any teacher before deciding to participate.

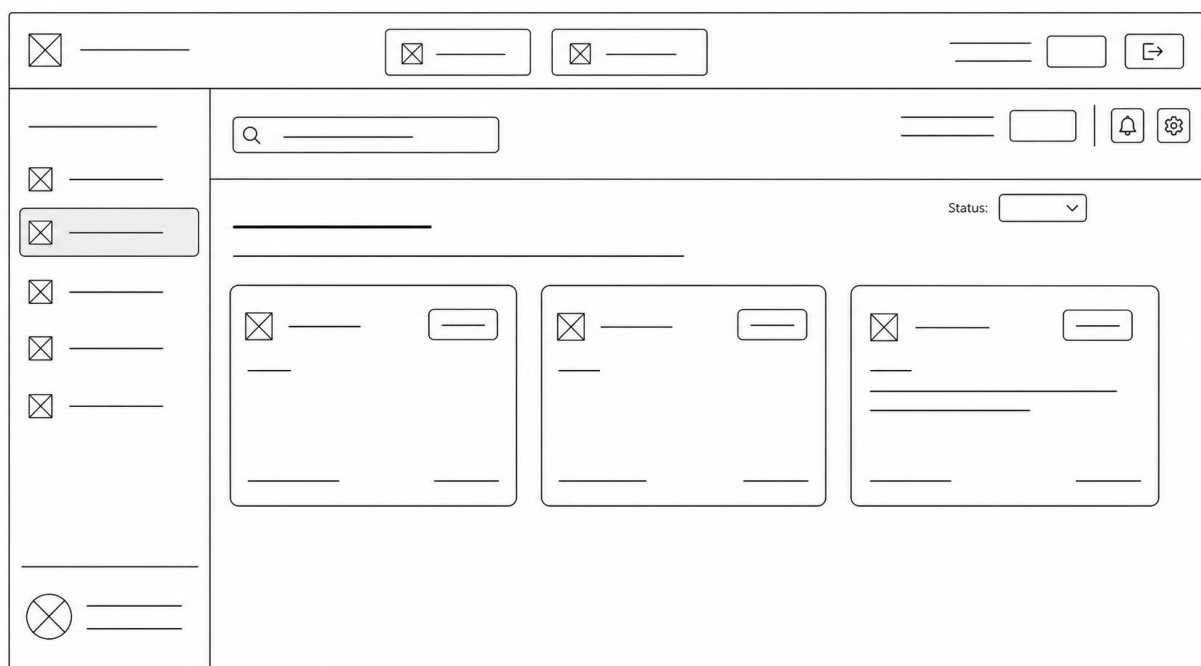


Figure 12.5 — Available Activities Screen (Student).

Reward Points History Screen

This screen supports the Student's Rewards History function (Section 6.3). The main panel presents the student's complete RP transaction history as a table, with columns for the activity or reason, category, date, and RP amount, and a status pill on each row distinguishing RP earned from RP used or deducted. A top navigation bar retains quick access to the Dashboard and Leaderboard, and the left sidebar highlights the current section. The table format allows the student to review individual RP transactions in chronological order and verify how their total RP was accumulated.



Figure 12.6 — Reward Points History Screen (Student).

Internal Marks Screen

This screen supports the Student's Internal Marks Module function (Section 6.3). The main panel opens with a page title and summary line, followed by a details card showing two side-by-side fields for the current marks tier and the RP range associated with it. Below this, a highlighted progress panel indicates the additional RP required to reach the next internal-marks tier, using a progress indicator. A further card beneath presents supporting detail on how the marks were derived. The top navigation bar and left sidebar match the rest of the Student module, keeping the internal marks view consistent with the Dashboard and other student-facing screens.

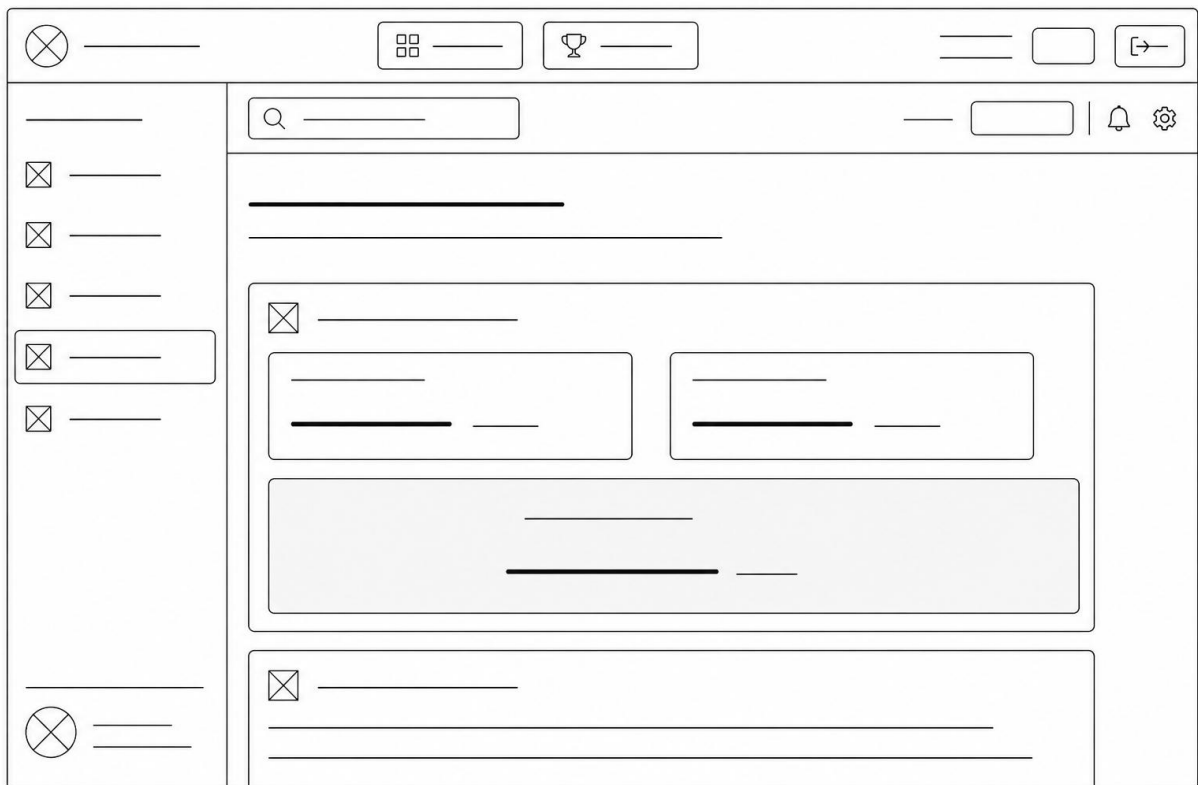


Figure 12.7 — Internal Marks Screen (Student).

Student Help Desk Screen

This screen supports the Student's Student Queries function (Section 6.3). The main panel is split into two areas. The left area contains the query submission form, with a subject field, a multi-line message field, and a send action for routing the query to the teacher responsible for the relevant category (or to Admin if uncategorized). The right area lists the student's previously raised queries as stacked cards, each showing a short summary and a button to view the full response. The left sidebar retains navigation to the student's other sections, with the help desk entry highlighted. This layout keeps query submission and the student's response history visible on the same screen.

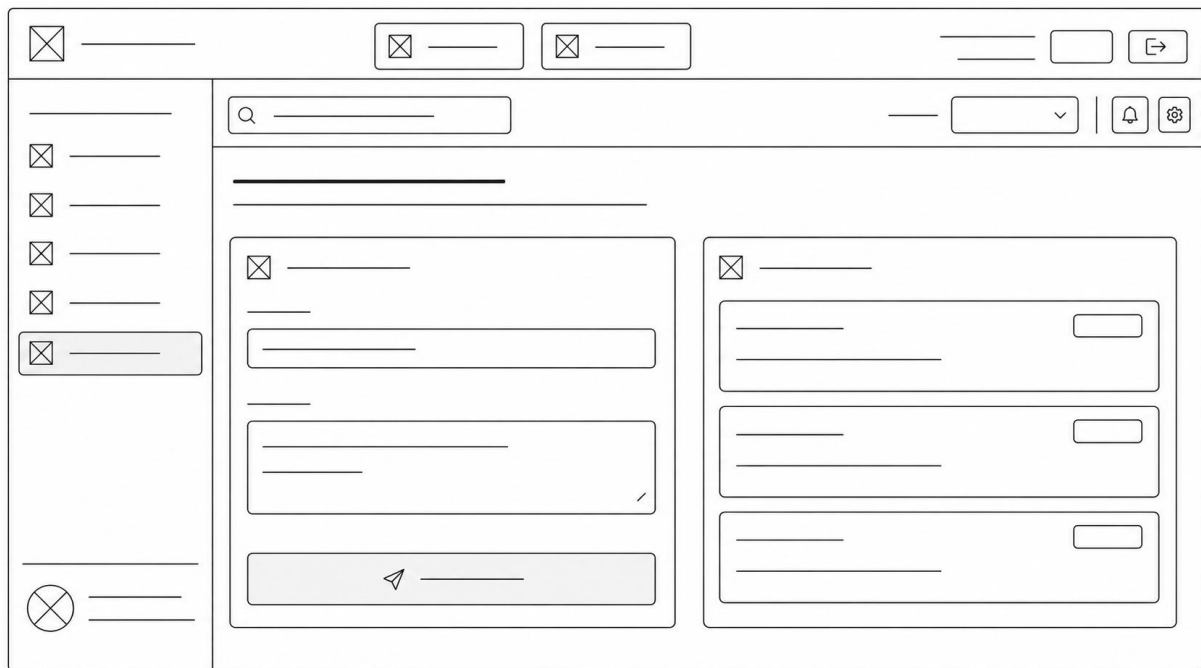


Figure 12.8 — Student Help Desk Screen (Student).

13. Future Enhancements

- Direct integration with the NPTEL portal for automated certification verification.
- Mobile native applications for Android and iOS.
- Configurable RP-to-internal-marks conversion policies per department.
- Additional teacher categories or sub-categories as institutional activities expand.
- Parent/guardian-facing read-only portal for performance visibility.

14. Conclusion

The Smart Reward Point Management and Redemption System provides an efficient, transparent, and role-segmented solution for managing student activities, reward points, and academic performance. The Teacher role — subdivided into Faculty 1, Faculty 2, and Faculty 3 categories — distributes activity management closer to the domain experts responsible for each category, while the Admin retains complete oversight through monitoring and reporting.

By integrating role-based access control, automation, analytics, and communication features, the system enhances student engagement, reduces administrative overhead, and offers a scalable, modern solution suitable for educational institutions.